

Research flow and what each step actually established

Experiment 1

Data & convention validation

QC established; released 2D vs provided 3D
self-consistent to 3.9977 mm (median)



Experiment 2

Focal sensitivity of absolute depth

a 5 % focal error moves the hand by about
33 mm in this working-distance regime



Experiment 3

Existing calibration baselines

distortion-aware modelling roughly halves the
error (16.3102 % -> 9.546 %), but ~10 % remains



Experiment 4

Multi-frame use & bias analysis

98 % of the residual is per-camera bias;
8-64 frames do not remove it



Current bottleneck

a stable per-camera / per-view bias,
not frame count and not frame choice



Next: CAM-EXP-005

why does each view have its own bias?
(scene / background / viewpoint cues)



Later

CAM-EXP-006 hand cues, then a single
sealed external confirmation

*dashed = planned, not yet performed. The +/- 5 % focal target is a working target derived from Experiment 2
in this working-distance regime, not a universal requirement.*